Time spent:0 sec

QID:[illegible]

Subject:
Biology

System:
Digestion and Excretion

Research into how the human body digests and further metabolizes ingested foods is critical to understanding human health processes and diseases, such as obesity and diabetes. However, researchers interested in studying the human digestive system must overcome many obstacles, including ethical constraints in using human and animal subjects, the high cost of experiments, and the lack of standardized conditions across human subjects. Because of these obstacles, researchers began designing and using *in vitro* digestive system models in the 1990s (Figure 1).

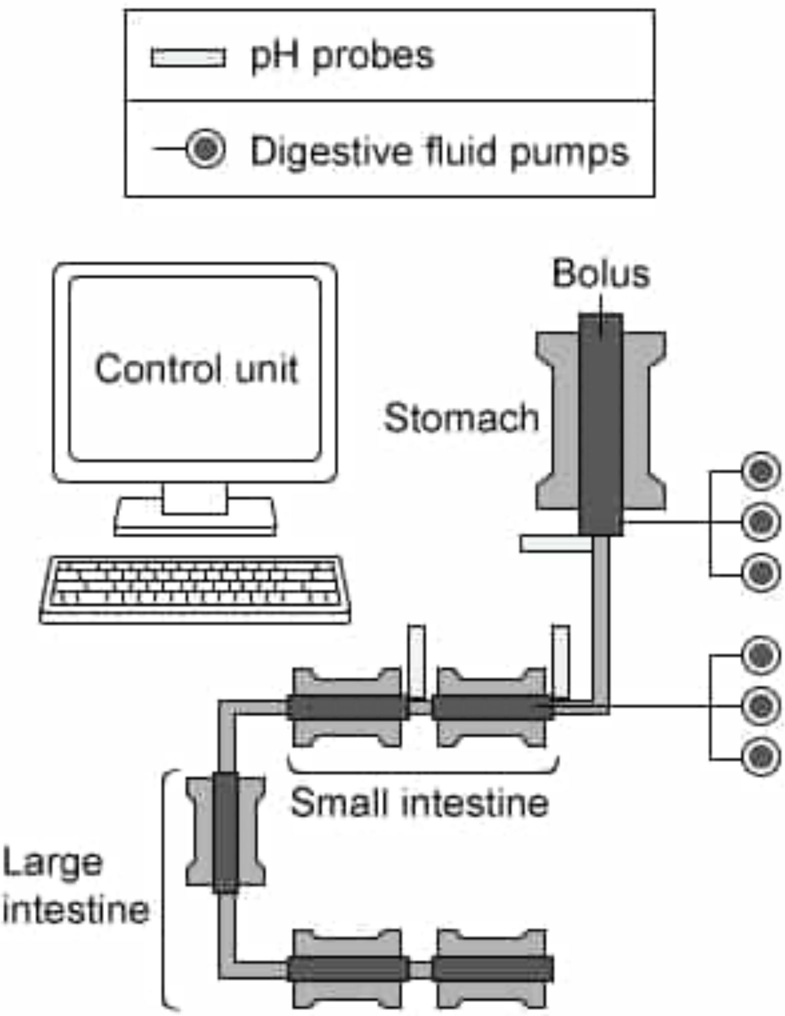


Figure 1 Example of *in vitro* digestive system model

There are two basic *in vitro* models of the human digestive system: static models and dynamic models, both of which have advantages and disadvantages. Static models, which make use of a series of glass chambers that model digestive organs, are more cost effective and practical for large sample sizes; however, they do not mimic the mechanical force or dynamic conditions produced in humans. While dynamic models are more costly, these model types come closer to approximating *in vivo* conditions, including mechanical, dynamic, and/or chemical conditions.

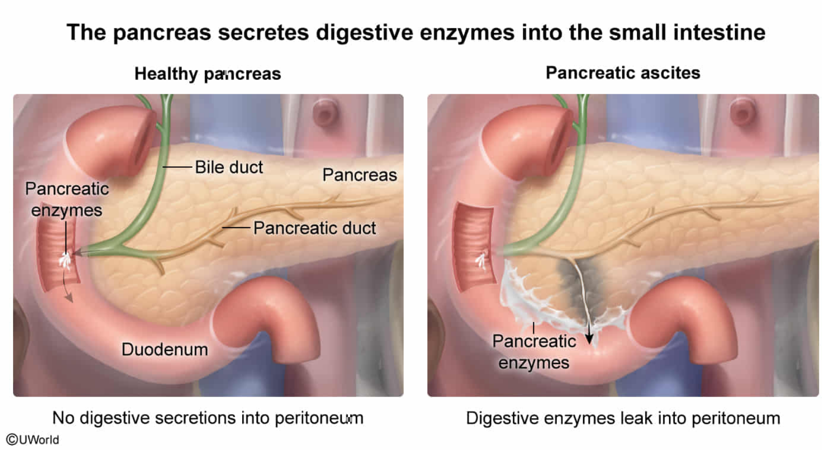
Regardless of model type, researchers employing *in vitro* digestive system models use a complex system of computer controls to adjust for variables such as pH, transit time of food particles, and digestive secretions.

Although the pancreas is located posterior to the peritoneal cavity, pancreatic ascites is a condition in which pancreatic secretions leak into the peritoneum. The best explanation for this condition would be leakage of:

- A. endocrine pancreas hormones as they are delivered to the small intestine. [5%]
- B. endocrine pancreas hormones as they are delivered to the blood supplying the digestive system. [19%]
- ✔ C. pancreatic enzymes as they are delivered to the small intestine. [62%]
- D. pancreatic enzymes as they are delivered to the blood supplying the digestive system. [12%]

Correct
62% answered correctly

Explanation:



The peritoneum lines the abdominal cavity and is made of two tissue layers. The first layer is known as the **parietal layer**, which lines the abdominal wall. The **visceral layer**, in contrast, covers many of the organs of the abdominal cavity. The **peritoneal cavity** is the space between these two layers. It is described as a potential space, or an area between two structures that are adjacent and may be pressed against one another.

This question states that the pancreas is located posterior to the peritoneal cavity; however, in pancreatic ascites, pancreatic secretions leak into the peritoneum. The endocrine pancreas secretes hormones involved in regulation of blood glucose into the bloodstream, while the exocrine pancreas secretes digestive enzymes and bicarbonate into the small intestine to assist in digestive processes and to neutralize the acidity of chyme.

Therefore, the best explanation for leakage of pancreatic secretions into the peritoneum during pancreatic ascites is that **pancreatic enzymes** leak as they are delivered to the duodenum of the **small intestine**.

(Choices A and B) The endocrine pancreas secretes hormones to the bloodstream that help regulate blood glucose; however, the endocrine pancreas is *not* directly involved in the process of digestion.

(Choice D) While pancreatic ascites involves leakage of pancreatic enzymes, these enzymes leak as they are delivered to the small intestine via the common bile duct, *not* to the blood supplying the digestive system.

Educational objective:
The peritoneum is made of two tissues layers, the parietal layer and the visceral layer, and lines the abdominal cavity. The peritoneal cavity is the potential space between these two layers.

Time spent:0 sec	**QID**:403885
Subject: Biology	**System:** Digestion and Excretion

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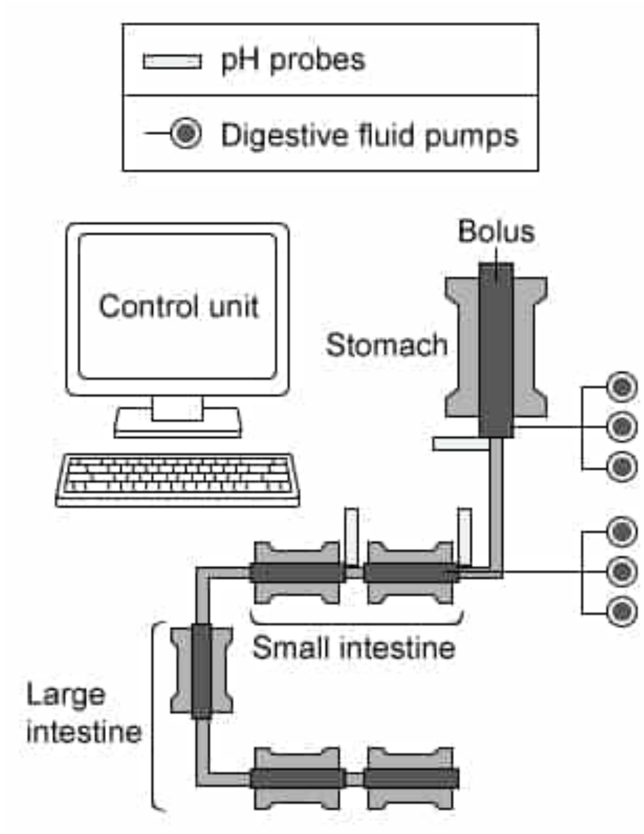


Figure 1 Example of *in vitro* digestive system model

There are two basic *in vitro* models of the human digestive system: static models and dynamic models, both of which have advantages and disadvantages. Static models, which make use of a series of glass chambers that model digestive organs, are more cost effective and practical for large sample sizes; however, they do not mimic the mechanical force or dynamic conditions produced in humans. While dynamic models are more costly, these model types come closer to approximating *in vivo* conditions, including mechanical, dynamic, and/or chemical conditions.

Regardless of model type, researchers employing *in vitro* digestive system models use a complex system of computer controls to adjust for variables such as pH, transit time of food particles, and digestive secretions.

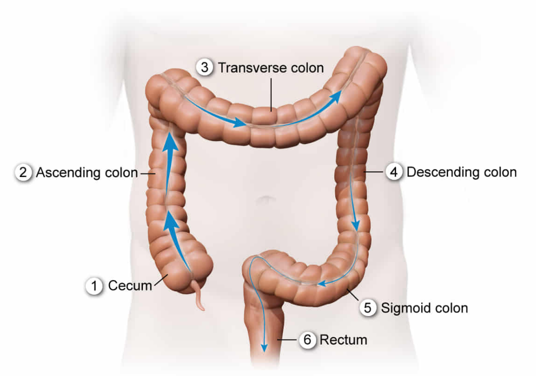
Some static *in vitro* digestive system models include stirring mechanisms with adjustable speeds to mimic the peristaltic waves in the large intestine. In the proximal large intestine, peristaltic contractions occur at a greater frequency than in the distal large intestine. Which series shows the anatomical order, from most frequent to least frequent, of peristaltic contraction frequency in the components of the large intestine that the static *in vitro* digestive system model should mimic?

- A. Ascending colon, transverse colon, descending colon, sigmoid colon, cecum [18%]
- B. Cecum, sigmoid colon, descending colon, transverse colon, ascending colon [7%]
- C. Cecum, transverse colon, ascending colon, descending colon, sigmoid colon [7%]
- ✔ D. Cecum, ascending colon, transverse colon, descending colon, sigmoid colon [66%]

Correct
66% answered correctly

Explanation:

Large intestine anatomy



As undigestible materials move through the large intestine, peristaltic contraction speed decreases.

The final portion of the gastrointestinal tract is the large intestine, a wide tube consisting of three subdivisions, the cecum, colon, and rectum. The colon is further subdivided into four sections: the ascending colon, transverse colon, descending colon, and sigmoid colon, from proximal to distal. In the large intestine, water and electrolytes are absorbed from undigestible material to form solid feces.

This question states that some static *in vitro* digestive system models include stirring mechanisms with adjustable speeds to mimic the peristaltic waves of contraction that occur in the large intestine to move material along this portion of the gastrointestinal tract. In the proximal large intestine, peristaltic contractions occur at greater frequency than in the distal large intestine. Therefore, the order of peristaltic contraction frequency, from most frequent to least frequent, that *in vitro* models should mimic is **cecum, ascending colon, transverse colon, descending colon, sigmoid colon**.

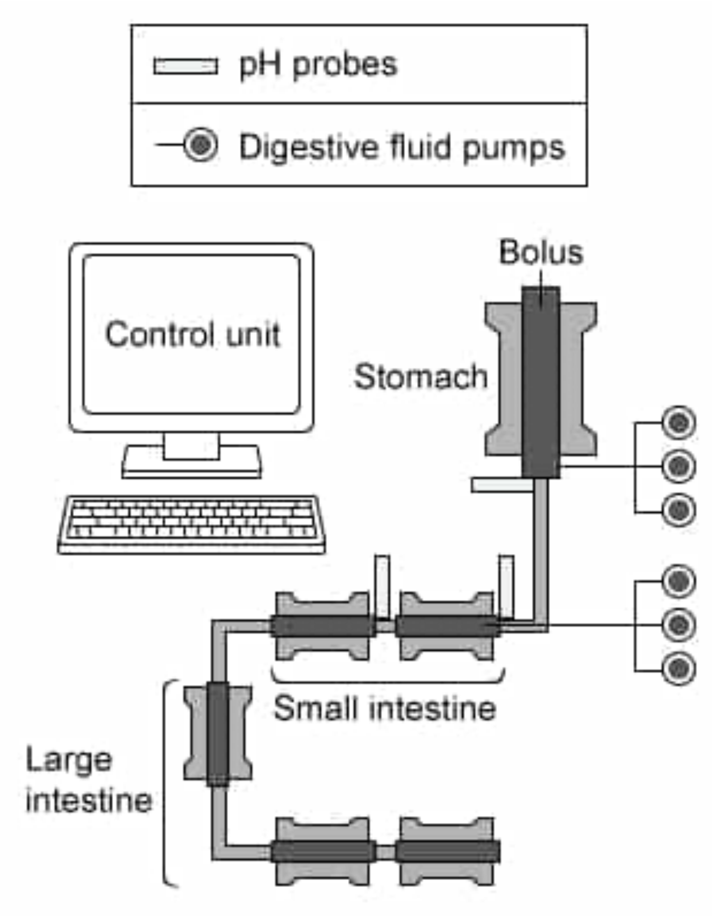
(Choices A, B, and C) These options represent *incorrect* proximal-to-distal ordering of large intestine components. Therefore, *in vitro* models with components in these orders would not correctly represent the *in vivo* large intestine.

Educational objective:
The large intestine, comprising the cecum, colon, and rectum, is responsible for absorbing water and electrolytes from undigestible material to form solid feces.

Time spent: 0 sec
Subject: Biology

QID:403886
System: Digestion and Excretion

Research into how the human body digests and further metabolizes ingested foods is critical to understanding human health processes and diseases, such as obesity and diabetes. However, researchers interested in studying the human digestive system must overcome many obstacles, including ethical constraints in using human and animal subjects, the high cost of experiments, and the lack of standardized conditions across human subjects. Because of these obstacles, researchers began designing and using *in vitro* digestive system models in the 1990s (Figure 1).



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Regardless of model type, researchers employing *in vitro* digestive system models use a complex system of computer controls to adjust for variables such as pH, transit time of food particles, and digestive secretions.

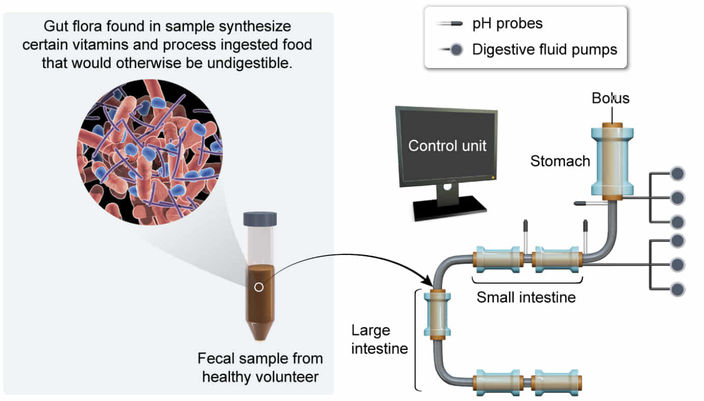
Some *in vitro* digestive system models include fecal matter obtained from healthy volunteers in the large intestine chambers. What is the most likely reason for the addition of this fecal matter?

- A. To confirm that gut flora metabolize only nutrients prior to digestive contents entering the large intestine chambers [11%]
- B. To determine the effect of gut flora from healthy volunteers on digestive contents within the chambers [61%]
- C. To test how much water is absorbed by the digestive system chambers [20%]
- D. To determine the effect of pathogenic bacteria on digestive processes inside the chambers [6%]

Correct
62% answered correctly

Explanation:

Effect of gut flora on *in vitro* digestive system models



The addition of fecal matter to *in vitro* digestive system models allows determination of the effect of gut flora on digestive contents.

The large intestine contains many bacterial species (ie, gut flora) that aid digestive processes. Some of these bacteria can process ingested foods that would be undigestible otherwise (eg, certain carbohydrates) into molecules (eg, short-chain fatty acids) that can then be absorbed by the body and used for energy. Other bacterial species in the large intestine synthesize certain vitamins.

This question states that some *in vitro* digestive system models include fecal matter obtained from healthy volunteers. This fecal matter would contain some of the bacterial flora found naturally in the human gut. Therefore, the inclusion of fecal matter would allow researchers to **determine the effect** of this **gut flora** on **digestive contents** in the model.

(Choice A) The flora of the large intestine metabolizes materials undigestible by the host; therefore, the addition of solid fecal matter from volunteers would *not* allow confirmation that gut flora metabolizes only nutrients prior to digestive contents entering the large intestine chambers.

(Choice C) One role of the large intestine is to absorb water and electrolytes from undigestible materials. However, the addition of *outside* materials to the *in vitro* model would *not* allow an accurate test of how much water is absorbed by the digestive system.

(Choice D) The question states that the fecal matter was obtained from healthy volunteers; therefore, it is unlikely that pathogenic bacteria would be found in the fecal matter. Consequently, this approach is unlikely to aid in determining the effect of pathogenic bacteria on digestive processes.

Educational objective:

Bacterial species found in the large intestine perform functions important to digestive health, including vitamin synthesis and aiding in the digestion of certain ingested foods that would otherwise be undigestible.

Time spent:0 sec	**QID**:403887
Subject:	**System:**
Biology	Digestion and Excretion